SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF MATHEMATICS
21MAB204T - PROBABILITY AND QUEUEING THEORY
Academic Year: 2025 - 2026 (Even Semester)

UNIT - IV : TUTORIAL SHEET – 10

S. No.	Problems
1.	A departmental store has a single cashier. During the rush hours, customers arrive at the rate of 20 customers per hour. The average no. of customers that can be processed by the cashier is 24 per hour. Calculate the following (i) What is the probability that the cashier is idle? (ii) What is the average number of customers in the queueing system. (iii) What is the average time a customer spends in the system. (iv) What is the average number of customers in the queue? (v) What is the average time a customer spends in the queue, waiting for service?
2.	If people arrive to purchase cinema tickets at the average rate of 6 per minute at a one man counter, and it takes an average of 7.5 seconds to purchase a ticket. If a person arrives 2 minutes before the picture starts and it takes exactly 1.5 minutes to reach the correct seat after purchasing the ticket, (i) can he expect to be seated for the start of the picture? (ii) What is the probability that he will be seated for the start of the picture? (iii) How early must he arrive in order to be 99% sure of being seated for the start of the picture?
3.	Customers arrive at a one man barber shop according to a Poisson process with a mean inter arrival time of 20 minutes. Customers spend an average of 15 minutes in the barber chair. If an hour is used as the unit of time, (i) What is the probability that a customer need not wait for a hair cut? (ii) What is the expected number of customers in the barber shop and in the queue? (iii) How much time can a customer expect to spend in the barbershop? (iv) Find the average time that a customer spends in the queue? (v) What is the probability that there will be more than 6 customers? (vi) What is the probability that there will be 6 or more customers waiting for service? (vii) What is the probability that the waiting time in the (a) system (b) queue, is greater than 12 minutes?
4.	Customers arrive at a watch repair shop according to a Poisson process at a rate of one per every 10 minutes and the service time is an exponential random variable with mean 8 minutes. (i) Find the average number of customers in the shop. (ii) Find the average time a customer spends in the shop. (iii) Find the average number of customers in the queue. (iv) What is the probability that the server is idle?
5.	An automatic car wash facility operates with only one bay. Cars arrive according to a poisson distribution with a mean of 4 cars per hour and may wait in the facility's parking lot if the bay is busy. If the service time for all cars is constant and equal to 10 minutes, determine (i) Mean number of customers in the system L_s. (ii) Mean number of customers in the system L_q. (iii) Mean waiting time of a customer in the system W_s. (iv) Mean waiting time of a customer in the system W_q.
6.	The arrivals at the counter in a bank occur in accordance with a Poisson process at an average rate of 8 per hour. The duration of service of a customer has an exponential distribution with a mean of 6 minutes. Find the probability that an arriving customer (i) Has to wait (ii) Finds 4 customers in the system (iii) Has to spend less than 15 minutes in the bank.